

JACOB HORNE

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EDUCATION

University of California, Irvine

June 2027

B.S. Computer Science, Campuswide Honors Collegium, Dean's Honor List

GPA: 3.91

Coursework: Data Structures & Algorithms, Computer Architecture, Software Engineering, System Design, Distributed Systems, Discrete Math, Computational Statistics, Linear Algebra, Data Mining, C++ Programming, AI

EXPERIENCE

Software Development Intern, SENS Psychology

Sep 2025 - Jan 2026

- Built a full-stack **Next.js/Node.js, TypeScript**, and **PostgreSQL** ticketing system that automated patient intake, call routing, and workflow tracking, improving response speed for a **30+** member front desk team.
- Designed **Prisma/Zenstack** data models supporting threaded messaging, task assignment, and lifecycle management with **React** and **shadcn/ui**, reducing manual triage efforts by **40%**.
- Automated patient and staff notifications via **Microsoft Azure**, reducing manual email workloads by **60%**.

Software Engineer Intern, CrowdQuant

Jun 2025 - Aug 2025

- Developed a full-stack **Python, FastAPI**, and **React** web platform for user access and ML model management, integrating authentication with **Supabase Auth** and JSON Web Tokens (**JWT**).
- Built PyTorch training and inference pipelines for **ResNet18** and **MobileNetV2** models, improving stock-trend forecasting accuracy by **10%** and reducing training time by **30%**.
- Utilized **Runpod** cloud development for scalable training, reduced overhead, and accelerated model-tuning.

Software Developer, CareTech @ UCI

Sep 2025 - Present

- Building an AI-powered nutrition analytics backend using **Python/FastAPI**, performing image segmentation and meal analysis with **OpenCV** and **YOLO**-based food detection, managed collaborative development using **Git/GitHub**.
- Trained **TensorFlow/PyTorch** models for calorie and meal prediction, integrated with **PostgreSQL** data layer.
- Implemented responsive **React** and **TailwindCSS** dashboard and containerized backend services with **Docker**.

Research Assistant, UCI Digital Learning Lab

Dec 2025 - Present

- Developing PapyrusAI, **multi-agent LLM** systems for student-facing educational applications and support devices.
- Analyzing student interaction data using **NLP** and **ML** to evaluate reasoning quality and usage patterns.

PROJECTS

[Github Onboarding Agent](#)

Sep 2025 - Dec 2025

- Built an AI agent that ingests any GitHub repo and generates structured project overviews, component summaries, and file-level explanations to help new contributors quickly understand unfamiliar codebases.
- Implemented **RAG** pipelines with **LangChain**, **Hugging Face** embedding models, **Pinecone** vector database.
- Developed a **FastAPI** backend and **Streamlit UI** powered by **LangGraph** multi-step reasoning and tool calling, enabling interactive repo exploration and automatic summaries of new commits and pull requests.

[SF Crime Mapper](#)

Mar 2025 - Apr 2025

- Built a **React** and **Mapbox** dashboard to visualize neighborhood-level crime predictions served from a **FastAPI** backend with an integrated, interactive map of SF neighborhoods.
- Engineered reusable **JavaScript** hooks and state logic to integrate **REST** endpoints for dynamic map updates.
- Linked the UI to **scikit-learn**, **XGBoost**, and **Prophet** predictive models in **Python**, surfacing dynamic crime-risk forecasts through the dashboard, visualizing predicted and historical neighborhood crime on the map.

[SMS Spam Detector](#)

Nov 2024 - Dec 2024

- Built ML-based classifier in **Python** using scikit-learn to detect phishing and spam SMS messages.
- Applied natural language processing (**NLP**) with tokenization, **TF-IDF vectorization**, & stopword removal.
- Designed simple **React, HTML/CSS** frontend, designed for seniors after Aunt was targeted by an SMS scam.

TECHNICAL SKILLS

Languages: Python, Javascript/Typescript, C/C++, SQL, Java, HTML/CSS, R

Frameworks: FastAPI, Next.js, React, Node.js

ML: PyTorch, TensorFlow, scikit-learn, XGBoost, Prophet, Numpy, pandas, Matplotlib

Tools: Docker, Git, PostgreSQL, Render, OpenCV, Supabase

AI/LLM: LangChain, LangGraph, RAG, Hugging Face, Pinecone, OpenAI API